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Classical trajectory and statistical adiabatic channel study of the dynamics of capture and unimolecular bond fission. VI. Properties of transitional modes and specific rate constants $k(E, J)$

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Transitional modes in simple unimolecular bond fission and in the reverse recombination reactions are characterized quantitatively by statistical adiabatic channel (SACM) and classical trajectory (CT) calculations. Energy E - and angular momentum J -specific numbers of open channels (or activated complex states) $W(E, J)$ and capture probabilities $w(E, J)$ are determined for a series of potentials such as ion–dipole, dipole–dipole, and various model valence potentials. SACM and CT treatments are shown to coincide under classical conditions. Adiabatic as well as nonadiabatic dynamics are considered. The dominant importance of angular momentum couplings is elaborated. A sequence of successive approximations, from phase space theory neglecting centrifugal barriers $E_0(J)$, via phase space theory accounting for centrifugal barriers $E_0(J)$, toward the final result, expressing the effects of the anisotropy of the potentials by specific rigidity factors $f_{\text{rigid}}(E, J)$, is described. This approach emphasizes the importance to characterize the employed potentials by their centrifugal barriers $E_0(J)$. The derived specific rigidity factors $f_{\text{rigid}}(E, J)$ are consistent with previously calculated thermal rigidity factors $f_{\text{rigid}}(T)$. The present approach properly accounts for angular momentum conservation and, at the same time, facilitates the calculation of specific rate constants $k(E, J)$ and falloff curves for unimolecular bond fission and the reverse radical recombination reactions. © 2002 American Institute of Physics. [DOI: 10.1063/1.1496463]

I. INTRODUCTION

Statistical rate theories can be formulated comparably easily when the considered processes are characterized by rigid activated complexes. In this case, partition functions and/or numbers of states of the activated complex are approximated by rigid rotor-harmonic oscillator models. The situation changes for loose activated complexes where “conserved modes” and “transitional modes” have to be distinguished, where centrifugal barriers appear at varying interreactant distances, and where complicated angular momentum coupling phenomena arise. This situation is typical for simple unimolecular bond fission and the reverse capture processes. In the present series of articles^{1–5} we have undertaken a systematic analysis of the relevant properties of “transitional modes” for a variety of reaction systems. We have employed two approaches, the statistical adiabatic channel model (SACM)^{6–8} for studying quantum effects under conditions where the dissociation/association dynamics is adiabatic, and classical trajectory (CT) calculations for studying the transition between adiabatic and sudden dynamics under classical conditions;⁹ the two approaches become equivalent when the conditions for classical and adiabatic

dynamics both are fulfilled (for the effects of rovibrationally nonadiabatic dynamics in the quantum range, see Refs. 10 and 11). The SACM/CT approach has the advantage that it provides a state-specific dynamical treatment of the bond fission/capture process which, with the sacrifice of some fine details, is simpler than a complete scattering treatment. However, it characterizes the dynamical properties of individual transitional modes in such a way that they can easily be implemented into an overall statistical description of the complete rate process.

The properties of the transitional modes depend on the relevant interfragment potential energy surface, its minimum energy path (MEP), and its anisotropy. Previously we have considered ion–dipole potentials,^{1,8,12} ion–quadrupole potentials,² dipole–dipole potentials,³ model valence potentials for capture of atoms by linear molecules⁴ and for capture of linear molecules by linear molecules.⁵ In addition, *ab initio* potentials for the O+NO,^{13–15} H+O₂,^{16,17} and O+OH¹⁸ association processes were employed. A systematic investigation of the dependence of *thermal* capture (or high pressure recombination) rate constants on molecular parameters was made with the model potentials treated in parts I–V.^{1–5} The comparison with *ab initio* potentials demonstrates to what extent predictions can be made without having access to *ab initio* results.

Energy E - and angular momentum (quantum number J) *specific* rate properties have not yet been studied systematically for the model potentials: this is the aim of the present work. The following strategy is employed.

At first, numbers of activated complex states (or num-

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bers of open channels) $W(E, J)$ for the transitional modes are derived neglecting centrifugal barriers $E_0(J)$; ^{19,20} these quantities are denoted by $W_0(E, J)$. The angular momentum coupling, preventing a separation of transitional modes in $W_0(E, J)$, is particularly manifest.

Second, the centrifugal barriers $E_0(J)$ are calculated along the MEP neglecting the anisotropy of the potential. At the same time, the maximum J (denoted by $J_{\max}$) is obtained for which the centrifugal barriers can be overcome. Both $E_0(J)$ and $J_{\max}(E)$ are crucial quantities characterizing the potential and entering kinetic parameters at various places. Accounting for the centrifugal barriers reduces $W_0(E, J)$ to the number of activated complex states in phase space theory (PST) which we denote by $W^{\text{PST}}(E, J)$. The ratio of $W^{\text{PST}}(E, J)$ and $W_0(E, J)$ determines the capture probability $w(E, J)$ in PST, i.e.,

$$w^{\text{PST}}(E, J) = W^{\text{PST}}(E, J) / W_0(E, J). \quad (1.1)$$

We demonstrate that $w^{\text{PST}}(E, J)$ in a simple way can be expressed by the centrifugal barriers $E_0(J)$.

Third, accounting for the anisotropy of the potential further reduces $W(E, J)$. We express this additional dynamical constraint by E - and J -specific rigidity factors defined by

$$f_{\text{rigid}}(E, J) = W(E, J) / W^{\text{PST}}(E, J). \quad (1.2)$$

In general, $f_{\text{rigid}}(E, J)$ is a complicated function depending on all details of the potential. However, in some cases $f_{\text{rigid}}(E, J)$ can be expressed by simple functions of $(J/J_{\max})^2$. We, therefore, choose similar representations for all types of interaction potentials.

Finally, numbers of states $W(E, J)$ for transitional modes are convoluted with contributions from conserved modes. Some preliminary results on $f_{\text{rigid}}(E, J)$ have been published before for model potentials ^{8,21,22} and for the $\text{O} + \text{NO}$, $\text{H} + \text{O}_2$, and $\text{O} + \text{OH}$ systems. ^{14,16,18} The present article completes our systematic general analysis.

On the basis of the described calculations one can obtain a clear picture of the properties of “activated complex modes”. Following SACM, ^{6–8} the “conserved modes” often approximately can be identified with the internal modes of the separated fragments. In PST, the “transitional modes” are external free rotors of the separated fragments which, however, are coupled by angular momentum constraints. ^{23–29} It is, therefore, well known that one cannot identify transitional modes with unconstrained “active rotors,” or to divide transitional modes into “active” and “inactive” rotors, such as this is sometimes done in naive applications of Rice–Ramsperger–Kassel–Marcus theory. Instead, transitional modes correspond to “special modes” with interesting new properties. They are neither anharmonic oscillators, nor free or conventional hindered rotors such as observed in spectroscopy. We illustrate this by looking at valence potentials. Expressing the anisotropy of these potentials by the ratio of the looseness parameter α and the Morse parameter β of the potential, ⁶ simplified SACM ⁷ for transitional modes led to $W(E, J=0) \propto E^x$ with energy-independent exponents x which approach unity for $\alpha/\beta \rightarrow 0$ and 0.5 for $\alpha/\beta \geq 0.5$. A hindered rotor-transitional mode would have led to energy-dependent exponents x . Standard valence potentials with $\alpha/\beta \approx 0.5$ ³⁰ in

this approximation correspond to transitional modes which approach angular momentum-constrained free rotors having, however, effective rotational constants larger than those of the free fragment rotors. ³¹

A consequence of energy-independent exponents x in $W(E, J) \propto E^x$ would be temperature-independent thermal rigidity factors $f_{\text{rigid}}(T)$. In this case, the rate constants would have temperature dependencies such as given by PST. As a matter of fact, only weakly temperature-dependent $f_{\text{rigid}}(T)$ have been encountered in the CT calculations for thermal capture rates from parts I–V ^{1–5} (for valence potentials, only as long as $\alpha/\beta = 0.5$). The present more systematic analysis reveals finer details of the behavior, particularly with respect to the E and J dependencies of $W(E, J)$ and $f_{\text{rigid}}(E, J)$. We believe that the considered model potentials can well serve as a reference for comparison with *ab initio* results. When *ab initio* potentials are not available, in addition they provide a realistic basis for predictions.

II. SPECIFIC RATE CONSTANTS $k(E, J)$, NUMBERS OF OPEN CHANNELS $W(E, J)$, AND CAPTURE PROBABILITIES $w(E, J)$

Within the framework of statistical unimolecular rate theory, the specific rate constants $k(E, J)$ for unimolecular reaction are expressed by

$$k(E, J) = W(E, J) / h \rho(E, J) \quad (2.1)$$

with the rovibrational densities of states $\rho(E, J)$ such as calculated with Refs. 7, 32–35. It has to be emphasized that the problem of anharmonicity contributions is far from being solved, see the examples discussed in Refs. 35–40. One should also remember that specific rate constants $k(E, J)$ are known to fluctuate over large ranges, ^{15,41} which is due to the fluctuations of the factor $1/h \rho(E, J)$ in Eq. (2.1).

We emphasize that the present treatment, by using Eq. (2.1), remains statistical even though $W(E, J)$ is determined by dynamical methods such as SACM/CT calculations. Separating conserved and transitional modes, $W(E, J)$ is obtained by convolution

$$W(E, J) \approx \int_0^E W_{\text{tr}}(E - E_i, J) [dW_{\text{cons}}(E_i, J) / dE_i] dE_i \quad (2.2)$$

of the numbers of open channels of the transitional $[W_{\text{tr}}(E, J)]$ and of the conserved modes $[W_{\text{cons}}(E, J)]$. Determining $W_{\text{cons}}(E, J)$ by state-counting for the separated fragments, see Eq. (2.2), in the following we focus attention on $W_{\text{tr}}(E, J)$ only and, for simplicity, we drop the subscript tr.

III. $W(E, J)$ FOR ADIABATIC CAPTURE IN ION–PERMANENT DIPOLE AND DIPOLE–DIPOLE POTENTIALS

In this section, the ion–permanent linear dipole and linear dipole–linear dipole systems are treated as representative examples of our approach. These systems have particularly simple properties such that our strategy can be illustrated most clearly.

The ion–linear dipole potential in center-of-mass (c.m.) coordinates has the form

$$V(r, \gamma) = -q\mu_D \cos \gamma / r^2 \quad (3.1)$$

with the ionic charge q , the dipole moment μ_D , the c.m. interfragment distance r , and the angle γ between the rotor axis and the line connecting the centers of mass of the fragments. In PST, $\cos \gamma$ is replaced by unity. Approximating the total angular momentum (quantum number J) by the orbital angular momentum (quantum number l), centrifugal barriers $E_0(J)$ are determined as maxima of the potential $V(r, J) = -q\mu_D/r^2 + J(J+1)\hbar^2/2\mu r^2$ with the reduced mass μ . The result is trivial, i.e.,

$$E_0(J) = 0 \quad \text{for } J \leq J_{\max} \quad (3.2)$$

with

$$J_{\max}(J_{\max} + 1) = 2\mu q\mu_D / \hbar^2, \quad (3.3)$$

and $E_0(J) = \infty$ for $J > J_{\max}$. As a consequence of Eq. (3.2), the capture probability in PST, $w^{\text{PST}}(E, J)$, in this case is equal to unity for $J \leq J_{\max}$ and zero for $J > J_{\max}$. The number of states

$$W_0(E, J) = \sum_l \sum_j 1 \quad (3.4)$$

of a linear rotor (quantum number j) and an orbiting ion (quantum number l) with a total angular momentum (quantum number J) is easily derived with the constraint $|J - j| \leq l \leq J + j$, giving⁷

$$W_0(E, J) = (2J + 1)(j_{\max} + 1) - J(J + 1) \quad (3.5)$$

at $0 \leq J \leq j_{\max}$, and

$$W_0(E, J) = (j_{\max} + 1)^2 \quad (3.6)$$

at $J \geq j_{\max}$ where $j_{\max}$ is the largest j with $0 \leq B j(j + 1) \leq E$ (B =rotational constant of the rotor).

Having characterized $W_0(E, J)$, $E_0(J)$, $J_{\max}$, and $w^{\text{PST}}(E, J)$, we finally determine rigidity factors $f_{\text{rigid}}(E, J)$ by including the $\cos \gamma$ anisotropy of the potential. The explicit calculation of adiabatic channel potential curves and their maxima in Ref. 8 in first approximation led to

$$f_{\text{rigid}}(E, J) \approx 1 - (J/J_{\max})^2. \quad (3.7)$$

Small deviations were noticed at small J and approximated by^{19,20}

$$f_{\text{rigid}}(E, J) \approx 1 - (J/J_{\max})^2 - 0.064 \exp[-9(J/J_{\max})^2]. \quad (3.8)$$

It appears worth emphasizing that $J_{\max}$ and f_{rigid} in this case do not depend on the energy E .

In the present work, instead of quantum adiabatic channel calculations we have repeated the calculation of f_{rigid} by classical trajectory calculations. Trajectories were followed from separated fragments to small distances. The same criteria for “capture” of the trajectories were employed as in parts I–V. Initial conditions for the trajectories were randomly chosen from a uniform distribution in phase space with boundaries compatible with the considered E and J .

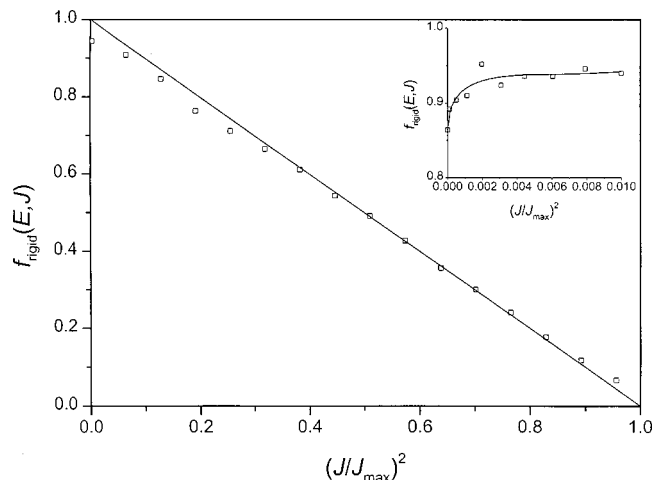


FIG. 1. Rigidity factors $f_{\text{rigid}}(E, J)$ for ion–permanent dipole capture [$\square$ =CT results; full line= $1 - (J/J_{\max})^2$ with $J_{\max}^2 \approx \hbar^2/2\mu q\mu_D$; insert: results at small $J/J_{\max}$ for effective mass $M=100$, see Eq. (3.9)].

Calculations were repeated until statistical scatter was smaller than 2%; in general this required about 10^4 trajectories for each $\{E, J\}$.

It was shown in Ref. 1 that two parameters govern the classical ion–linear dipole capture dynamics, an effective mass M and an effective anisotropy measure θ . For thermal capture, these parameters were given as $M = \mu q\mu_D B / \hbar^2 kT$ and $\theta = \alpha kT / 2\mu_D^2$ where α is the polarizability of the dipole rotor. In order to distinguish these quantities from the present case, we term them $M(T)$ and $\theta(T)$. The corresponding dimensionless quantities relevant for the present microcanonical situation are

$$M(E) = \mu q\mu_D B / \hbar^2 E \quad (3.9)$$

and $\theta(E) = \alpha E / 2\mu_D^2$. $M(T)$ and $M(E)$ are related to the Massey parameter $\xi = \sqrt{2M}$, defined in Ref. 9, which separates the ranges of adiabatic ($\xi > 1$) and nonadiabatic, sudden, dynamics ($\xi < 1$). With increasing E , a transition from adiabatic to sudden dynamics takes place which can well be localized via Eq. (3.9). In the present work, we do not further consider nonadiabatic dynamics in ion–dipole capture. Instead, in Sec. VI we only analyze nonadiabatic effects for capture in valence potentials (an analysis of nonadiabatic dynamics for thermal capture is given in parts I–V^{1–5}).

The results of our CT calculations of specific rigidity factors, for ion–permanent dipole capture, are illustrated in Fig. 1. The SACM results⁸ from Eqs. (3.7) and (3.8) are fully confirmed. In addition, however, a minor decline of f_{rigid} at $J \rightarrow 0$ is documented [in the insert of Fig. 1, for an effective mass of $M(E) = 100$] which corresponds to the transition to planar rotation. According to Eqs. (3.5) and (3.6), this transition at $J = j_{\max}$ is located at $(J/J_{\max})^2 = 1/2M$, i.e., at $(J/J_{\max})^2 = 0.005$ in Fig. 1. Neglecting the described deviations from Eq. (3.7), the product of Eqs. (3.5) and (3.6) with Eq. (3.7) specifies $W(E, J)$ for the ion–permanent dipole system.

The linear dipole–linear dipole potential is expressed in the form

$$V = \frac{\mu_{D_1}\mu_{D_2}}{r^3}[(\mathbf{d}_1\mathbf{d}_2) - 3(\mathbf{d}_1\mathbf{n})(\mathbf{d}_2\mathbf{n})] \quad (3.10)$$

with the dipole moments μ_{D_1} and μ_{D_2} of the two rotors, and the unit vectors $\mathbf{d}_1$, $\mathbf{d}_2$, and $\mathbf{n}$ along the dipolar axes and the line connecting the centers of mass. In this case, the effective mass $M(E)$ is defined by

$$M(E) = \mu B(\mu_{D_1}\mu_{D_2})^{2/3}/(\hbar^2 E^{2/3}) \quad (3.11)$$

with the average rotational constant of the two rotors

$$B = (B_1 + B_2)/2. \quad (3.12)$$

Neglecting the anisotropy, the centrifugal barriers analogous to the ion–dipole case are derived to be

$$E_0(J) = \{J(J+1)\hbar^2/[6\mu(\mu_{D_1}\mu_{D_2})^{2/3}]\}^3. \quad (3.13)$$

Inserting Eq. (3.13) gives $J_{\max}$, i.e.,

$$J_{\max}(J_{\max}+1) = 6\mu E^{1/3}(\mu_{D_1}\mu_{D_2})^{2/3}/\hbar^2. \quad (3.14)$$

The number of states $W_0(E, J)$ instead of Eq. (3.4) now is given by

$$W_0(E, J) = \sum_{j_1} \sum_{j_2} \sum_j \sum_l 1 \quad (3.15)$$

with the constraints $0 \leq B_1 j_1(j_1+1) \leq E$, $0 \leq B_2 j_2(j_2+1) \leq E - B_1 j_1(j_1+1)$, $|j_1 - j_2| \leq j \leq j_1 + j_2$, and $|J - j| \leq l \leq J + j$. The explicit expression for $W_0(E, J)$ is too cumbersome to be given here. However, in classical approximation it can conveniently be interpolated^{7,19} between

$$W_0(E, \text{small } J) = 2(2J+1)E^{3/2}[B_1^{1/2} + B_2^{1/2} - (B_1 + B_2)^{1/2}]/(3B_1 B_2) \quad (3.16)$$

and

$$W_0(E, \text{large } J) = E^2/(2B_1 B_2). \quad (3.17)$$

Having characterized the centrifugal barriers $E_0(J)$, by Eqs. (3.2) and (3.3) for the ion–linear dipole potential and by Eqs. (3.13) and (3.14) for the linear dipole–linear dipole potential, capture probabilities $w^{\text{PST}}(E, J)$ in PST, which account for centrifugal barriers, can be calculated. Following the definition of Eq. (1.1) and considering only the range where $J \approx l$ [i.e., the range of validity of Eqs. (3.6) and (3.17)], for the linear dipole–linear dipole case, one has

$$w^{\text{PST}}(E, J) = \{[E - E_0(J)]/E\}^2 = [1 - E_0(J)/E]^2; \quad (3.18)$$

on the other hand, for the ion–linear dipole system,

$$w^{\text{PST}}(E, J) = 1 - E_0(J)/E \quad (3.19)$$

applies which, with $E_0(J)=0$, leads to $w^{\text{PST}}(E, J)=1$ at $J \leq J_{\max}$, and $w^{\text{PST}}(E, J)=0$ at $J > J_{\max}$. Numerical results for more complicated potentials are illustrated in the following.

Accounting for the anisotropy of the potential, the CT results for the rigidity factors of the dipole–dipole system in Fig. 2 finally are represented as a function of $J/J_{\max}$. We find that

$$f_{\text{rigid}}(E, J) \approx 0.92[1 - (J/J_{\max})^2]^3 \quad (3.20)$$

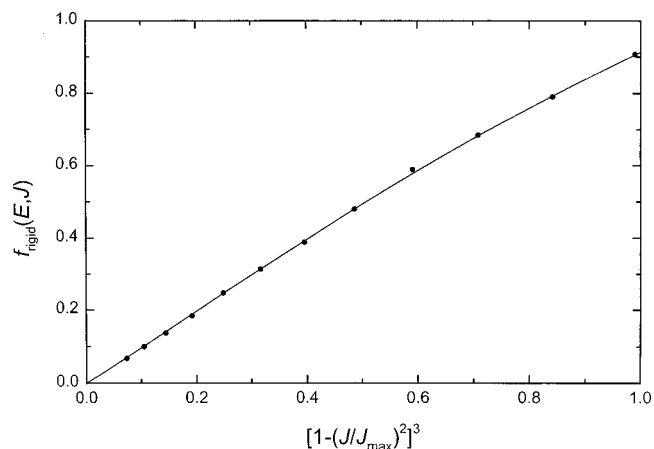


FIG. 2. Rigidity factors $f_{\text{rigid}}(E, J)$ for dipole–dipole capture without dispersion forces [■=CT results; $J_{\max}$ given by Eq. (3.14)].

gives an excellent approximation to the numerical results. One should note that in this case, through the energy dependence of $J_{\max}$, f_{rigid} is a function of E and J . Equations (3.13)–(3.18) finally lead to the complete $W(E, J)$.

It should be mentioned that thermal averaging over $W(E, J)$ leads to the same thermal capture rate constants as derived independently in Refs. 1 and 3. For the ion–linear dipole system, in this way a thermal rigidity factor of

$$f_{\text{rigid}}(T) \approx 0.50 \quad (3.21)$$

is obtained; likewise, for the linear dipole–linear dipole system, one derives

$$f_{\text{rigid}}(T) \approx 0.354. \quad (3.22)$$

IV. $W(E, J)$ FOR ADIABATIC CAPTURE IN GENERAL ION–DIPOLE AND DIPOLE–DIPOLE POTENTIALS

In this section we investigate the influence of additional isotropic contributions to the potentials of the ion–linear dipole and the linear dipole–linear dipole systems. Instead of Eq. (3.1), we consider

$$V(r, \gamma) = -q\mu_D \cos \gamma/r^2 - \alpha q^2/2r^4 \quad (4.1)$$

with the polarizability α ; instead of Eq. (3.10), we employ

$$V = \frac{\mu_{D_1}\mu_{D_2}}{r^3}[(\mathbf{d}_1\mathbf{d}_2) - 3(\mathbf{d}_1\mathbf{n})(\mathbf{d}_2\mathbf{n})] - \frac{c_6}{r^6} \quad (4.2)$$

with the dispersion constant c_6 . We introduce the parameters

$$c_1 = \alpha E/2\mu_D^2 \quad (4.3)$$

and

$$c_2 = c_6 E/(\mu_{D_1}\mu_{D_2})^2, \quad (4.4)$$

which measure the relative amounts of isotropy versus anisotropy of the potential. Obviously, the rigidity factors should be modified, being between the results from Sec. III for $c_1=0$ and $c_2=0$ and unity for $c_1 \rightarrow \infty$ and $c_2 \rightarrow \infty$.

The centrifugal barriers, for the ion–dipole system, now are

$$E_0(J) = 0 \quad \text{for } J(J+1) \leq 2\mu q \mu_D/\hbar^2 \quad (4.5)$$

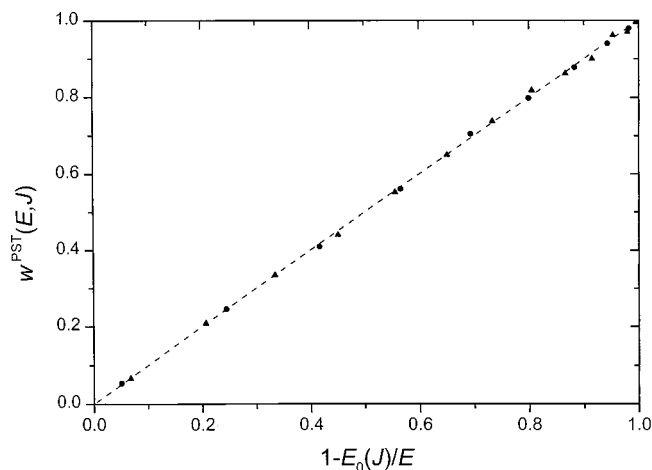


FIG. 3. Capture probabilities $w^{\text{PST}}(E, J)$ in PST for ion–permanent+induced dipole capture [symbols=CT result; $\bullet$: $c_1 = 1$ and $\blacktriangle$: $c_1 = 10$ with $c_1 = \alpha E/2\mu_D^2$, see Eq. (4.3)].

and

$$E_0(J) = [J(J+1)\hbar^2/2\mu q\mu_D - 1]^2(\mu_D^2/2\alpha) \quad (4.6)$$

for $J(J+1) \geq 2\mu q\mu_D/\hbar^2$. In order to demonstrate the quality of our CT calculations, in Fig. 3 we compare $w^{\text{PST}}(E, J)$ from these calculations with the analytical results from Eqs. (3.19), (4.5), and (4.6). In addition, after inverting Eqs. (4.5) and (4.6) to $J_{\text{max}}(E)$, Fig. 4 shows the rigidity factors as a function of $(J/J_{\text{max}})^2$. It should be mentioned that our previ-

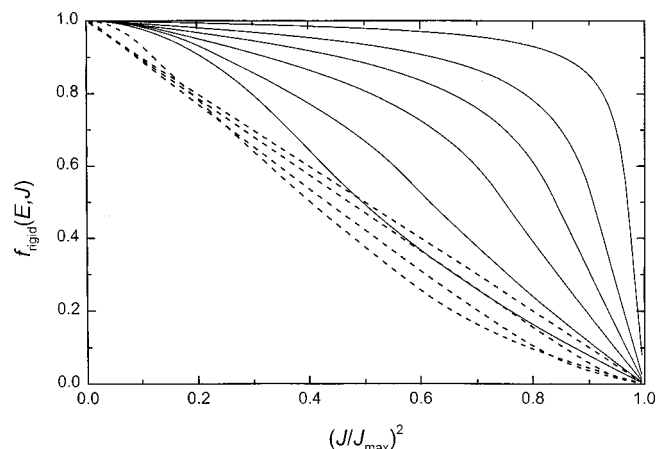


FIG. 4. Rigidity factors $f_{\text{rigid}}(E, J)$ for ion–permanent+induced dipole capture [dashed lines: $c_1 = 0, 0.001, 0.01, 0.1$ from top to bottom; solid lines: $c_1 = 1, 3, 10, 30, 100, 1000$ from bottom to top, $c_1 = \alpha E/2\mu_D^2$, see Eq. (4.3)].

ous quantum SACM results from Refs. 21 and 22 are well reproduced. The transition from the anisotropic permanent dipole to the isotropic induced dipole case is well documented. Exploiting the oscillator model of part III,³ the capture probabilities

$$w(E, J) = w^{\text{PST}}(E, J) f_{\text{rigid}}(E, J) \quad (4.7)$$

can also be derived directly. With the abbreviations $G_1 = J(J+1)\hbar^2/2\mu q\mu_D$, $a_1 = 1 - G_1$, and c_1 from Eq. (4.3), one obtains

$$w(E, J) = \begin{cases} 1 - G_1^2/4c_1 & \text{for } 0 \leq G_1 \leq G_0 = (1 + 4c_1)^{1/2} - 1 \\ [(a_1^2 + 12c_1)^{1/2} + a_1][(a_1^2 + 12c_1)^{1/2} - 2a_1]/4c_1 & \text{for } G_0 \leq G_1 \leq G_{00} = 1 + 2c_1^{1/2} \\ 0 & \text{for } G_1 \geq G_{00} \end{cases} \quad (4.8)$$

Figure 5 compares these analytical expressions for $w(E, J)$ with CT results [putting $J(J+1) \approx J^2$].

The linear dipole–linear dipole+dispersion potential of Eq. (4.2) is more difficult to handle analytically. We derived $J_{\text{max}}(J_{\text{max}}+1)\hbar^2/2\mu E^{1/3}(\mu_{D1}\mu_{D2})^{2/3}$

$$= 3 \frac{(1 + 8c_2)^{1/2} + 3 \left[\frac{(1 + 8c_2)^{1/2} - 1}{4c_2} \right]^{1/3}}{4} \quad (4.9)$$

Inversion of Eq. (4.9) gives $E_0(J)$. Representing $w^{\text{PST}}(E, J)$, $f_{\text{rigid}}(E, J)$, and $w(E, J)$ as a function of $(J/J_{\text{max}})^2$, Figs. 6–8 graphically illustrate our results. The pure linear dipole–linear dipole case is included as the limit $c_2 \rightarrow 0$.

V. $w(E, J)$ FOR ADIABATIC CAPTURE IN VALENCE POTENTIALS

In the following we consider dissociation/association processes of systems which are governed by valence potentials. In order to illustrate the general behavior, we use “standard forms” of these potentials like in Refs. 4 and 5. Indi-

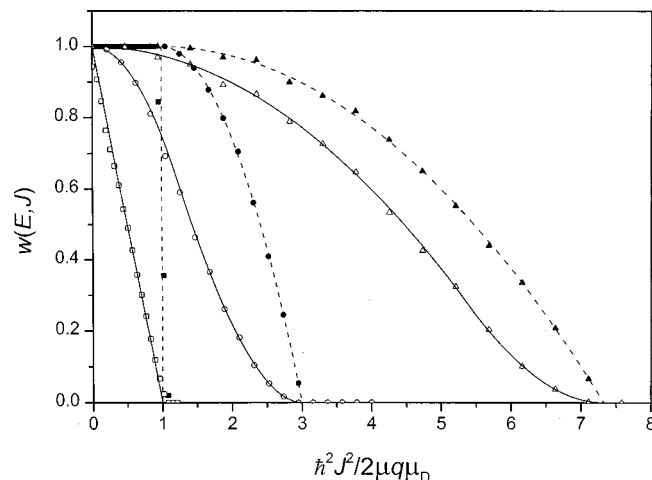


FIG. 5. Capture probabilities $w(E, J)$ for ion–dipole capture [CT results; closed symbols: PST; open symbols: anisotropic potential; $\blacksquare, \bullet, \blacktriangle$: $c_1 = 0, 1, 10$; $\square, \circ, \triangle$: $c_1 = \alpha E/2\mu_D^2$, see Eq. (4.3)].

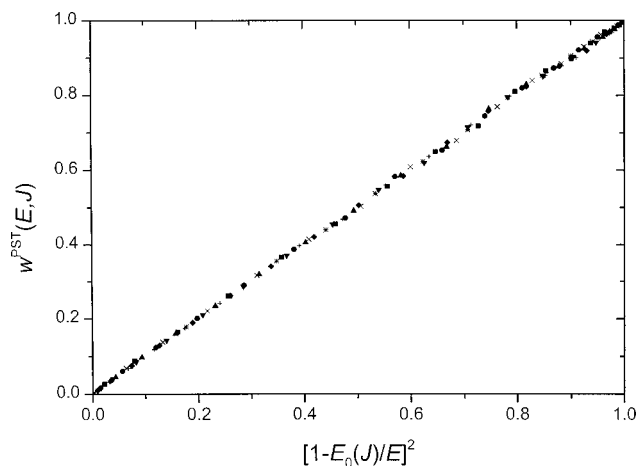


FIG. 6. Capture probabilities $w^{\text{PST}}(E, J)$ in PST for dipole–dipole capture [CT results for $c_2=0$ (■), 0.25 (●), 0.5 (▲), 1 (▼), 2 (◆), 4 (+), 8 (×), and 16 (*) with $c_2=c_6E/(\mu_{D_1}\mu_{D_2})^2$, see Eq. (4.4)].

vidual reaction systems, for which *ab initio* potentials are available, can be compared with the results from this simplified treatment. In this section, only adiabatic dynamics is considered; nonadiabatic situations are illustrated in Sec. VI. “Atom+linear rotor” systems forming linear or T-shaped adducts as well as “linear rotor+linear rotor” systems forming linear adducts are investigated.

Neglecting the anisotropy of the potential, we consider MEP potentials which, in c.m. coordinates, can well be approximated by simple Morse expressions of the form

$$V(r) = -2D \exp[-\beta(r-r_e)] + D \exp[-2\beta(r-r_e)] \quad (5.1)$$

with the Morse parameter β , the Morse dissociation energy D , and the equilibrium value r_e of the c.m. interfragment distance r . The effective mass M of the system, which corresponds to the square of the ratio between the characteristic times for the collision and for the rotation of the rotor fragment, now is given by

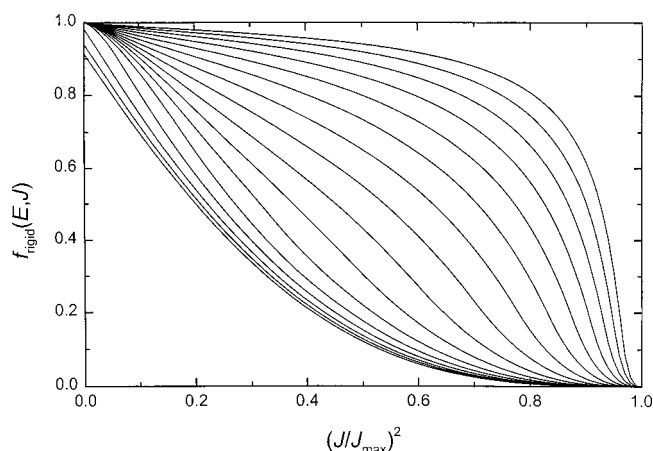


FIG. 7. Rigidity factors $f_{\text{rigid}}(E, J)$ for dipole–dipole capture ($c_2=0, 0.125, 0.25, 0.5, 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024$, and 2048 from bottom to top, with c_2 from Eq. (4.4)].

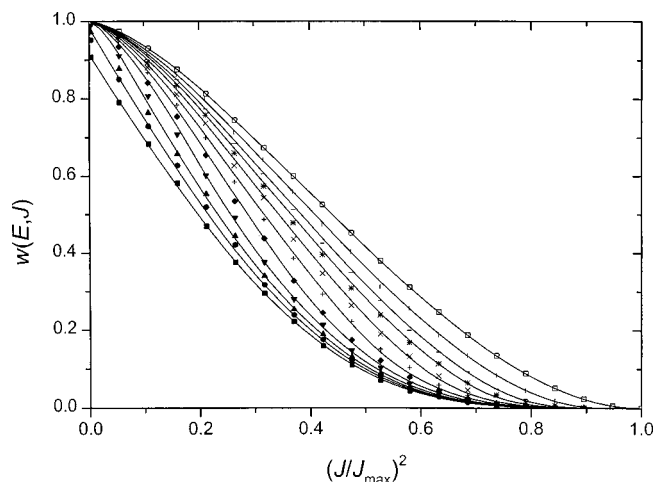


FIG. 8. Capture probabilities $w(E, J)$ for dipole–dipole capture (CT results for $c_2=0$ (■), 0.25 (●), 0.5 (▲), 1 (▼), 2 (◆), 4 (+), 8 (×), 16 (*), 32 (—), 128 (||), and ∞ (□), with c_2 from Eq. (4.4)).

$$M = \mu B / \hbar^2 \beta^2, \quad (5.2)$$

i.e., by an energy independent quantity. In the present section we only consider adiabatic situations with $M \gg 1$.

Expressing the centrifugal energies $E_0(J)$ in analytical form, for valence potentials, is not an easy task.⁷ Likewise, the maximum of J leading to capture at a given energy E takes a complicated form. Since the maximum of the sum of $V(r)$ from Eq. (5.1) and $J(J+1)\hbar^2/2\mu r^2$ is easily determined numerically, here we do not present analytical expressions for $E_0(J)$ and $J_{\text{max}}(E)$. We note, however, that $W_0(E, J)$ from Eqs. (3.5) and (3.6) for atom+linear rotor systems, and from Eqs. (3.16) and (3.17) (with the interpolations from Refs. 20 and 42) for linear rotor+linear rotor systems, are again the basis of the treatment. In addition, $w^{\text{PST}}(E, J)$ with the relevant $E_0(J)$ again is expressed by Eq. (3.19) for the former, and by Eq. (3.18) for the latter systems. This part of the treatment is the same, for the present model potential and for *ab initio* potentials, after the respective $E_0(J)$ have been derived, see, e.g., Ref. 14.

The rigidity factors $f_{\text{rigid}}(E, J)$, as in parts IV and V,^{4,5} are calculated with model anisotropies which are of simple $\cos \gamma$ form when a linear adduct is formed from atom+linear rotor fragments. In this case, the potential

$$V(r, \gamma) = -2D \exp[-\beta(r-r_e)] + D \exp[-2\beta(r-r_e)] + A_1[(1 - \cos \gamma)/2] \exp[-\beta(r-r_e)] \quad (5.3)$$

is employed⁴ which corresponds to a ratio $\alpha/\beta=0.5$ of the looseness and Morse parameters. Here, the r -dependent vibrational quantum $\hbar \omega(r)$ of the bending oscillator, corresponding to the transitional mode, is related to A_1 by⁴

$$\hbar \omega(r) = \hbar [(A_1/2)(2B/\hbar^2 + 1/\mu r_e^2)]^{1/2} \times \exp[-\beta(r-r_e)/2]. \quad (5.4)$$

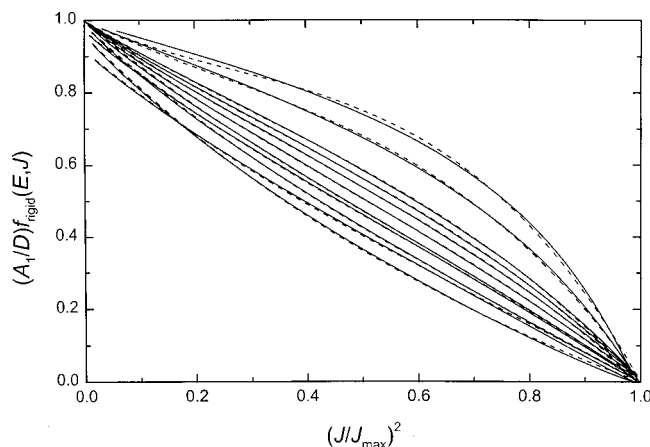


FIG. 9. Rigidity factors $f_{\text{rigid}}(E, J)$ for atom+linear rotor $\rightleftharpoons$ linear rotor valence systems (effective mass $M \rightarrow \infty$; $\beta r_e = 2.4$; $\lambda_0 = 10^{12}, 10^8, 10^5, 3 \times 10^4, 10^4, 3 \times 10^3, 10^3, 3 \times 10^2, 10^2, 30$ with $\lambda_0 = 2D \exp(\beta r_e)/E$; solid lines from oscillator model, dashed lines: fit to $f_{\text{rigid}} \propto [1 - z + az(1 - z)(z - b)]$ with $z = (J/J_{\text{max}})^2$; A_1/D =anisotropy amplitude, see Eq. (5.3); results for $A_1/D > 1$, see the text).

We have investigated the consequences of introducing the anisotropy term of the potential either by employing CT calculations or by using the modified harmonic oscillator model from Ref. 3. The results were very similar, depending largely on the anisotropy parameter $A_1/4D$ and on $(J/J_{\text{max}})^2$; a weaker dependence on the parameter $\lambda = (2D/E) \exp(\beta r_e)$ was also noticed. Figure 9 shows the results, for $A_1 > 4D$, which do not differ too much from the approximation

$$f_{\text{rigid}}(E, J) \approx (4D/A_1)[1 - (J/J_{\text{max}})^2]. \quad (5.5)$$

At the same time, w^{PST} besides Eq. (3.19) is also well approximated by

$$w^{\text{PST}} \approx 1 - (J/J_{\text{max}})^2, \quad (5.6)$$

which we do not illustrate here. Thermal averaging over Eqs. (5.5) and (5.6) leads to

$$f_{\text{rigid}}(T) \approx 8D/3A_1 \quad (5.7)$$

in close agreement with the direct thermal result of $f_{\text{rigid}} \approx 2\sqrt{2}D/A_1$ from Ref. 4.

Besides the “atom+linear rotor $\rightleftharpoons$ linear adduct” system, like in part IV⁴ we also have considered a valence system of the type “atom+linear rotor $\rightleftharpoons$ T-shaped adduct.” Centrifugal barriers $E_0(J)$, $J_{\text{max}}(E)$, numbers of states $W_0(E, J)$, and capture probabilities $w^{\text{PST}}(E, J)$ in PST obviously remain the same as for the linear adduct. Only the different anisotropy of the system results in different rigidity factors $f_{\text{rigid}}(E, J)$. Following Ref. 4, the anisotropic potential of Eq. (5.3) for a T-shaped adduct is changed into

$$V(r, \gamma) = -2D \exp[-\beta(r - r_e)] + D \exp[-2\beta(r - r_e)] + A_2[3(\cos^2 \gamma)/2] \exp[-\beta(r - r_2)]. \quad (5.8)$$

We have again determined f_{rigid} by CT calculations and by solution of the modified oscillator model from Ref. 5. The results are identical such as illustrated in Fig. 10. A satisfactory approximation of the obtained values, for $A_2 > 3D$, is given by

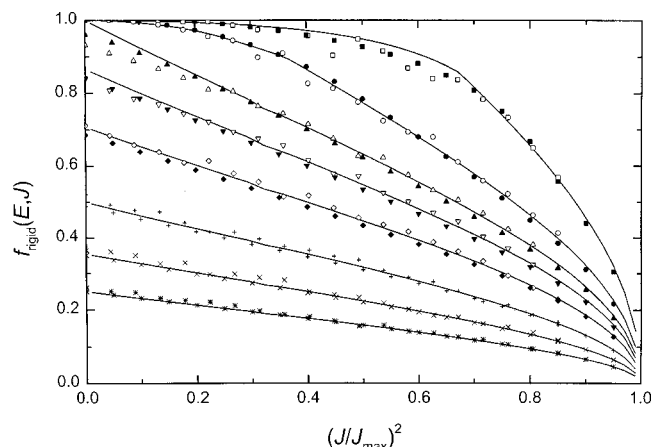


FIG. 10. Rigidity factors $f_{\text{rigid}}(E, J)$ for atom+linear rotor $\rightleftharpoons$ T-shaped adduct valence systems (CT results for $M \rightarrow \infty$; $\lambda_0 = 10^3$, see Fig. 9; $\beta r_e = 2$; $A_1/D = 1$ ($\square$), 2 ($\circ$), 3 ($\triangle$), 4 (∇), 6 ($\diamond$), 12 ($+$), 24 ($\times$), and 48 ($*$), open symbols=CT results, closed symbols=modified oscillator model from Ref. 5).

$$f_{\text{rigid}}(E, J) \approx (3D/A_2)^{1/2}[1 - (J/J_{\text{max}})^2]^{1/2}. \quad (5.9)$$

Thermal averaging over Eqs. (5.6) and (5.9), for $A_2 > 3D$, then again leads to good agreement with the direct result from Ref. 4, i.e., to

$$f_{\text{rigid}}(T) \approx \sqrt{2}(D/A_2)^{1/2}. \quad (5.10)$$

Finally, we have studied the “linear rotor+linear rotor $\rightleftharpoons$ linear adduct” system, following the treatment of part V⁵ and using a potential of the form

$$V(r, \gamma) = -2D \exp[-\beta(r - r_e)] + D \exp[-2\beta(r - r_e)] + A_3\{2 + [\mathbf{d}_1 \mathbf{d}_2 - 3(\mathbf{d}_1 \mathbf{n})(\mathbf{d}_2 \mathbf{n})]\} \times \exp[-\beta(r - r_e)]/4 \quad (5.11)$$

with the unit vectors $\mathbf{d}_1$, $\mathbf{d}_2$, and $\mathbf{n}$ defined as in Eq. (3.10). We have again only considered the large mass limit. Centrifugal barriers $E_0(J)$ and $J_{\text{max}}(E)$ are calculated numerically, leading to $w^{\text{PST}}(E, J)$ such as represented by Eq. (3.18). $W_0(E, J)$ as before is given by Eqs. (3.15)–(3.17) and the mentioned interpolation. The rigidity factors arising from the anisotropy term in Eq. (5.11) are illustrated in Fig. 11. Apart from the energy dependence of $J_{\text{max}}(E)$, there is only a small additional energy dependence such as documented in Fig. 11. This corresponds to similar observations in Fig. 9. A simple approximation of the results in Fig. 11, for $A_3/D > 4$, is given by

$$f_{\text{rigid}}(E, J) \approx (2D/A_3)^2[1 - (J/J_{\text{max}})^2]. \quad (5.12)$$

Thermal averaging leads to⁵

$$f_{\text{rigid}}(T) \approx 12.5(D/A_3)^2. \quad (5.13)$$

Equation (5.12) applies to small values of $(2D/A_3)^2$; for larger values, see Fig. 11. It should be emphasized that Eqs. (5.5), (5.9), and (5.12) are relatively crude approximations of our results presented in the figures. However, they appear satisfactory for many purposes. Furthermore, the transition to

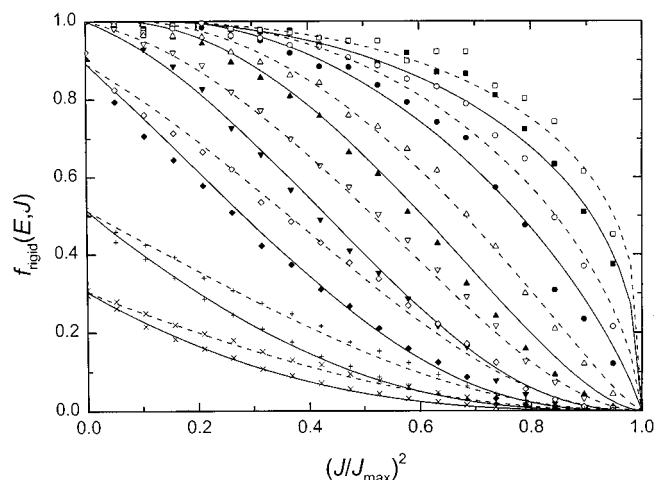


FIG. 11. Rigidity factors $f_{\text{rigid}}(E, J)$ for linear rotor+linear rotor $\rightleftharpoons$ linear adduct valence systems [$\lambda_0=300$ (full lines) and 10 000 (dashed lines), see Fig. 9; $\beta r_e=2$; CT results for $A_2/D=0, 0.5, 1, 2, 3, 4, 6$, and 8 from top to bottom].

isotropic potentials, when A_1 , A_2 , or A_3 become smaller than D , is not represented by these relations, see Figs. 10 and 11 instead.

The quantity A_3 , which characterizes the anisotropy of the potential, is related to the rotational constants of the two linear rotors and the corresponding vibrational frequencies of the transitional modes in the adduct. In our previous work, we have provided this relation only for the case of two identical rotors.⁵ In order to complete our study, in the Appendix of the present work we also derive the corresponding expressions for two nonidentical rotors forming various types of adducts.

VI. CAPTURE PROBABILITIES AND RIGIDITY FACTORS FOR NONADIABATIC DYNAMICS IN VALENCE POTENTIALS

The results of Secs. III–V all applied to adiabatic dynamics, characterized by large effective mass M . In the fol-

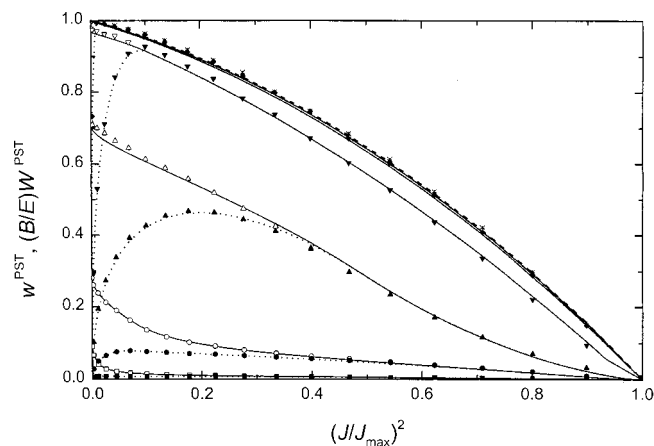


FIG. 12. Capture probabilities $w^{\text{PST}}(E, J)$ (full lines) and numbers of activated complex states $W^{\text{PST}}(E, J)$ (dotted lines) in PST for atom+linear rotor $\rightleftharpoons$ linear rotor valence systems ($\lambda_0=10^2$, see Fig. 9; CT results for effective masses $M=10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100$ from bottom to top).

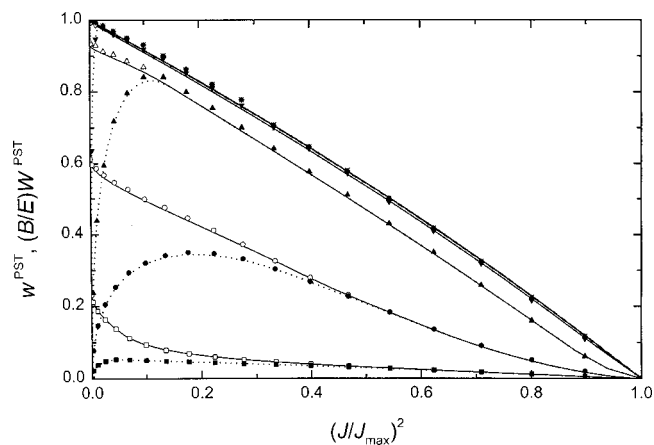


FIG. 13. As in Fig. 12, for $\lambda_0=10^6$.

lowing, for atom+linear rotor $\rightleftharpoons$ linear adduct systems in valence potentials, we inspect the effects of reducing M . In PST, reducing M leads to a transition from two-dimensional to one-dimensional planar rotation such as characterized by the transition between Eqs. (3.5) and (3.6). At the same time, the capture probability $w^{\text{PST}}(E, J)$ decreases compared to the large M limit. In addition, the rigidity factor changes such as described in the present section.

In Figs. 12 and 13 we illustrate the dependence of the capture probabilities $w^{\text{PST}}(E, J)$ on M . For $M \rightarrow 0$, there is a decrease of $w^{\text{PST}}(E, J=0)$ from unity to smaller values. The E and J dependencies of $w^{\text{PST}}(E, J)$ also deviate from the form characterizing $M \rightarrow \infty$, i.e., from Eq. (3.19). Figures 12 and 13 in addition include the numbers of activated complex states in PST, $W^{\text{PST}}(E, J)$. The range of the planar rotation, corresponding to Eq. (3.5), with decreasing M becomes increasingly broad. The contribution from Eq. (3.5), as well as the smaller values of $w^{\text{PST}}(E, J)$, with decreasing M increasingly reduce $W^{\text{PST}}(E, J)$ below the value obtained for $M \rightarrow \infty$.

Besides $w^{\text{PST}}(E, J)$, with decreasing M also the rigidity factors $f_{\text{rigid}}(E, J)$ change their character. In the limit

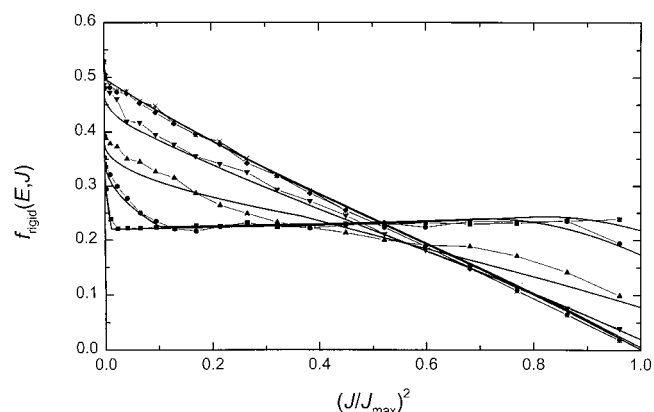
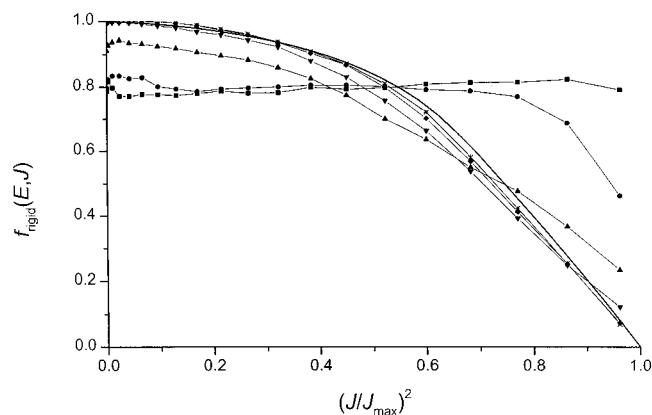


FIG. 14. Rigidity factors $f_{\text{rigid}}(E, J)$ for atom+linear rotor $\rightleftharpoons$ linear rotor valence systems [$\lambda_0=10^3$, see Fig. 9; $\beta r_e=2$; $A_1/D=8$, see Eq. (5.3); CT results for effective masses $M=10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100$ from bottom to top].

FIG. 15. As in Fig. 14, for $A_1/D=2$, see Eq. (5.3).

$M \rightarrow 0$ corresponding to the case of limiting nonadiabatic (sudden) dynamics, and for $A_1/2D > 1$ in the potential of Eq. (5.3), one has

$$f_{\text{rigid}}(J=J_{\text{max}}) \rightarrow 2D/A_1. \quad (6.1)$$

In addition, the J dependence is only weak such that Eq. (6.1) provides a good approximation for general J . The transition from adiabatic to sudden dynamics in general is of the type illustrated in Figs. 14 and 15. Apart from the energy dependence of $J_{\text{max}}(E)$, there is some additional weak energy dependence of the curves which is not shown here. It turns out that $f_{\text{rigid}}(E, J)$ does not depend on M for $(J/J_{\text{max}})^2$ near

to 0.5, at least for not too small energies. We do not give here analytical representations of our results for nonadiabatic dynamics. However, we mention that thermally averaged, nonadiabatic, rate constants have been documented in parts I–V.^{1–5}

VII. ADIABATIC AND AXIALLY NONADIABATIC CHANNEL CALCULATIONS

Whenever single (or small numbers of) channels, fine energy or fine angular momentum resolutions, or effects of open electronic shells are of importance, the classical trajectory calculations of Secs. III–V have to be replaced by explicit adiabatic channel calculations. Such calculations have been made for ion–dipole potentials of $\cos \gamma$ anisotropy in Ref. 8, for ion–quadrupole potentials of $\cos^2 \gamma$ anisotropy in Ref. 43, and for dipole–dipole potentials corresponding to the anisotropy of Eq. (5.11) in Ref. 44. The effects of open electronic shells have also been documented for ion–dipole,⁸ ion–quadrupole,¹² and dipole–dipole systems.⁴⁵ The transfer of these results to the present valence potentials is straightforward: while the radial potentials as well as the amplitudes of the r -dependent anisotropy potentials have to be scaled, the general channel pattern and the correlations between adduct and separated rotor states remain the same.

In the following, examples for “atom+linear rotor” and “linear rotor+linear rotor” valence potentials are given. SACM calculations for $W(E, J)$ are compared with quantized PST calculations for $W^{\text{PST}}(E, J)$ in Figs. 16–18. The

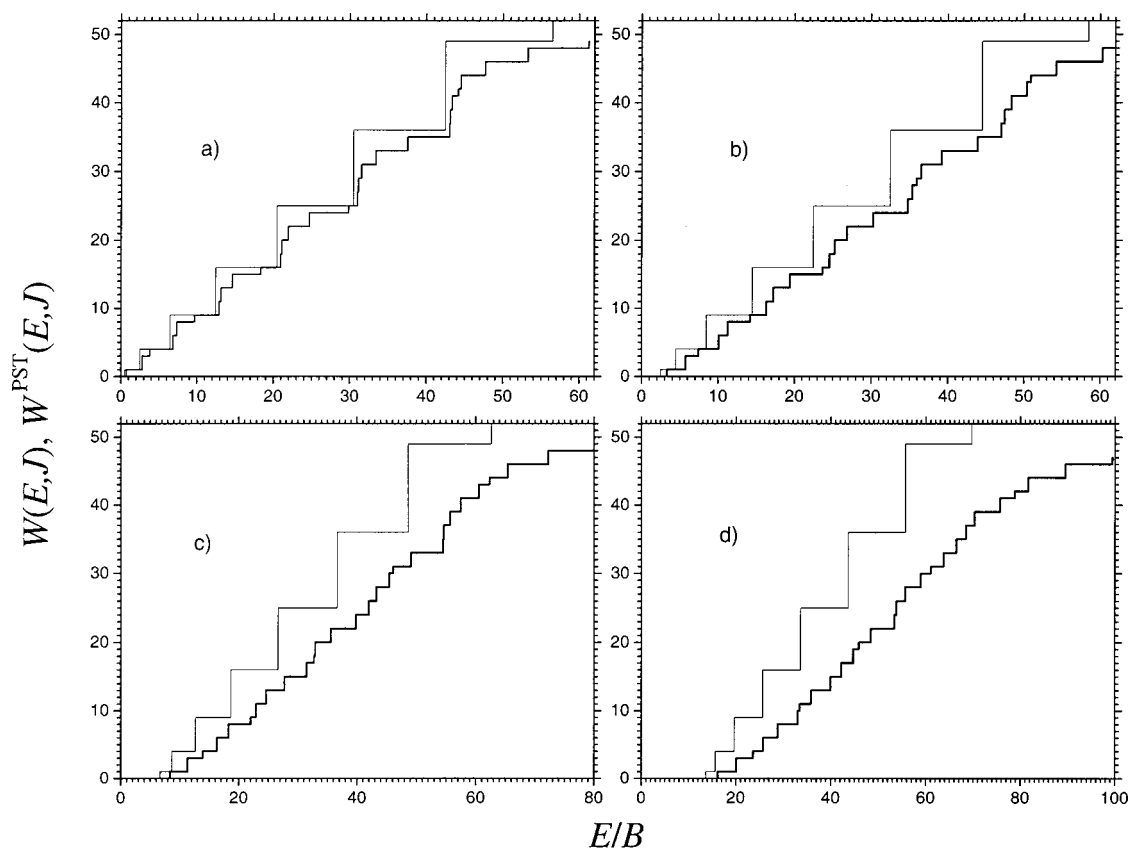
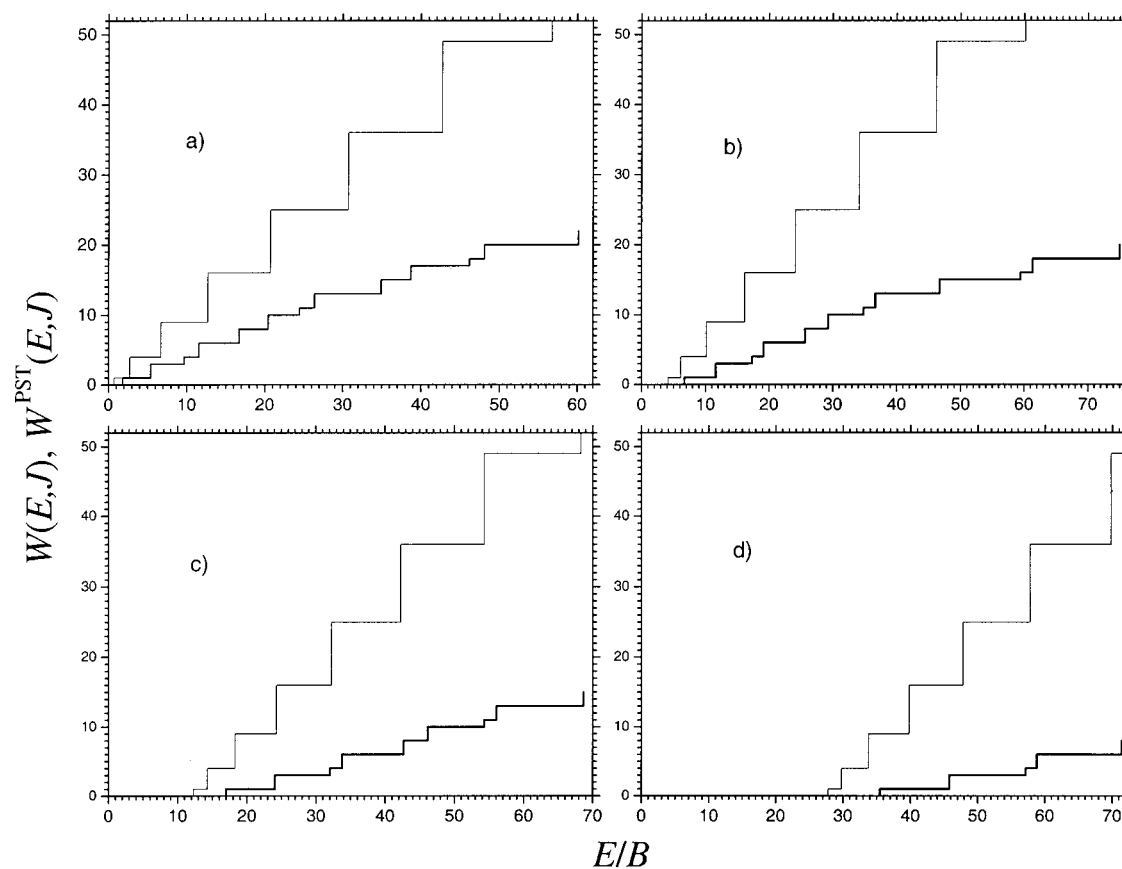
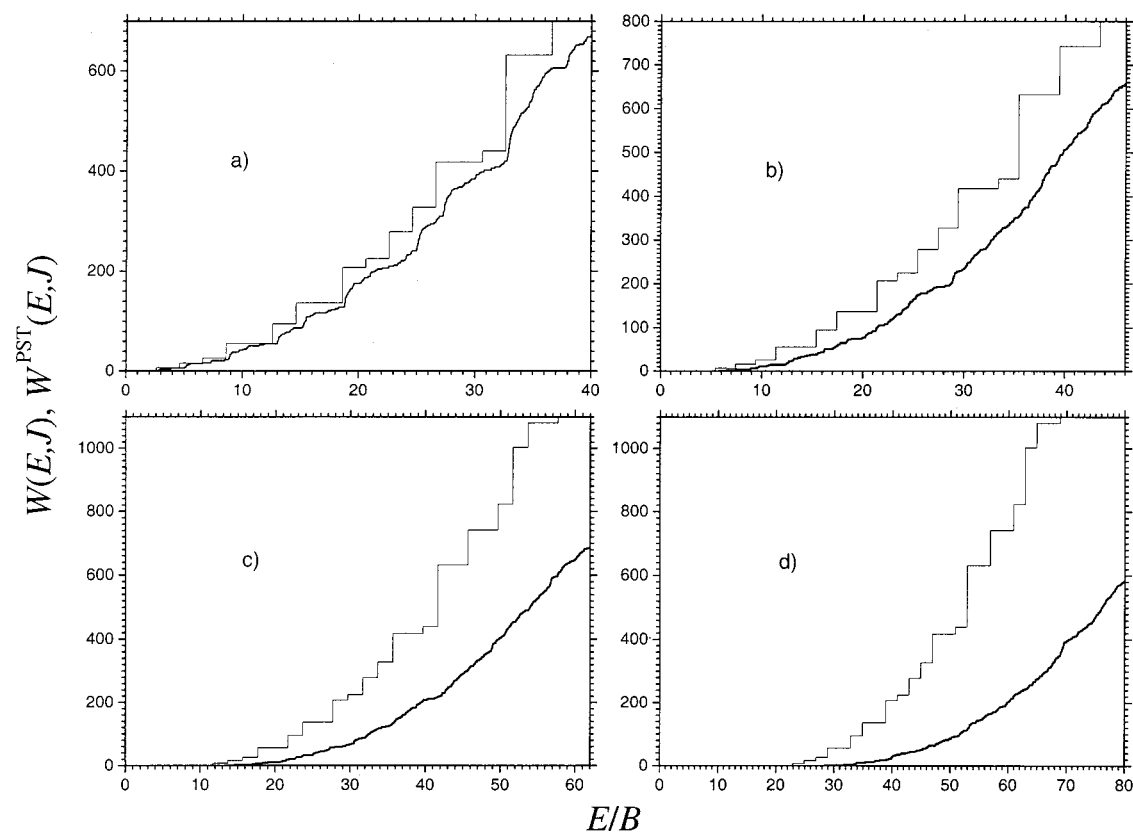


FIG. 16. Numbers of open channels, $W(E, J)$, and numbers of open channels in PST, $W^{\text{PST}}(E, J)$, for atom+linear rotor $\leftrightarrow$ linear rotor valence systems [adiabatic channel calculations, $A_1/D=4$; $\beta r_e=4$; $A_1/B=2000$; $\hbar J(2\mu\beta^{-2}B)^{-1/2}=10, 20, 30, 40$ for (a), (b), (c), (d), respectively].

FIG. 17. As in Fig. 16, $A_1/D=8$.FIG. 18. Numbers of open channels, $W(E, J)$, and numbers of open channels in PST, $W^{\text{PST}}(E, J)$, for linear rotor+linear rotor $\leftrightarrow$ linear rotor valence systems [adiabatic channel calculations, $A_2/D=4$; $\beta r_e=4$; $A_2/B=500$; $\hbar J(2\mu\beta^{-2}B)^{-1/2}=10, 20, 30, 40$ for (a), (b), (c), (d), respectively].

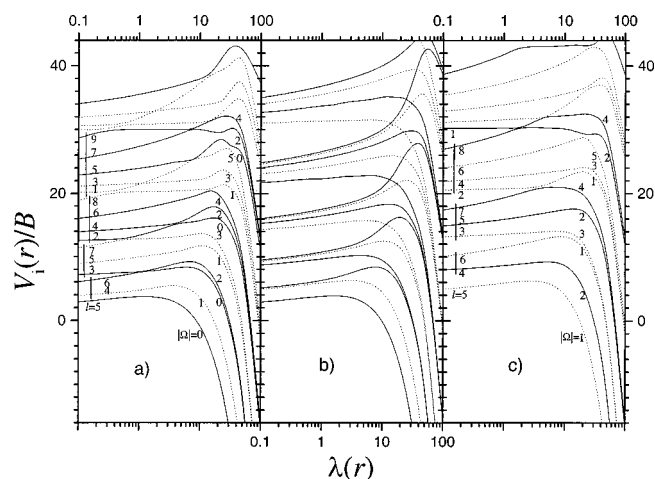


FIG. 19. Nonadiabatic channel potential curves $V_i(r)$ for atom+linear rotor $\rightleftharpoons$ linear rotor valence systems [potential from Eq. (5.3) with $A_1/D = 4$, $M = 0.02$, $\beta r_e = 2$, $\lambda(r) = \lambda_1 \exp(-\beta r)$, $\lambda_1 = A_1 \exp(\beta r_e)/2B = 10^6$; calculations for $J^\sigma = 5^+$ (panel a), $J^\sigma = 5^-$ (panel c) accounting for Coriolis coupling, and for $J = 5$ (panel b) neglecting Coriolis coupling, σ =parity; orbital quantum numbers l joined in rotational multiplets for separated atom+rotor (Ref. 10); vibrational momentum quantum numbers Ω for the adduct; dotted and full curves for odd and even bending vibrational quantum numbers, respectively].

irregular step patterns, both of $W(E, J)$ and $W^{\text{PST}}(E, J)$, result in irregular step functions of the specific rigidity factors $f_{\text{rigid}}(E, J) = W(E, J)/W^{\text{PST}}(E, J)$. On average, however, these $f_{\text{rigid}}(E, J)$ were shown to fluctuate precisely around the corresponding CT results from Sec. V, see the following. (As there are numerous accidental avoided crossings of the adiabatic channel potentials in the linear rotor–linear rotor case, see Ref. 43, diabatic curves were chosen whenever the splittings at the avoided crossings of the adiabatic channels were smaller than 0.1 B).

Since apart from the quantum step nature the SACM and CT results of the curves in Figs. 16–18 were quite similar, in

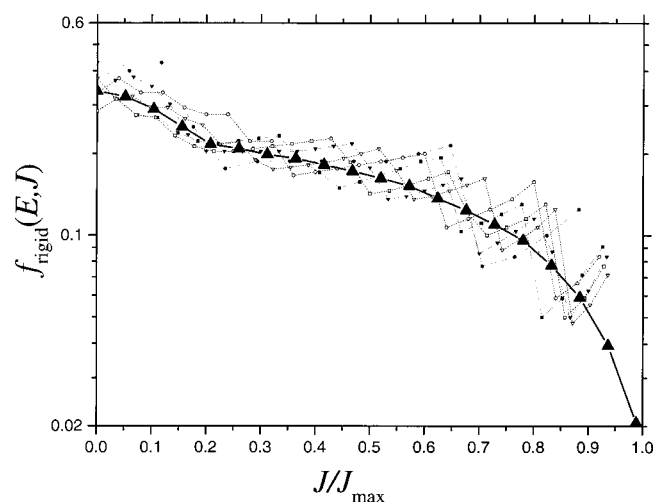


FIG. 20. Rigidity factors $f_{\text{rigid}}(E, J)$ for atom+linear rotor $\rightleftharpoons$ linear rotor valence systems [for parameters see Fig. 9; $\lambda_0 = 10^3$, $\beta r_e = 2$, $A_1/D = 16$, $M = 0.1$, $J_{\text{max}} = 4.29(E/B)^{1/2}$; $\blacktriangle$: CT calculations, nonadiabatic channel calculations including Coriolis coupling for $E/B = 20$ ($\bullet$), 40 ($\circ$), 45 ($\blacksquare$), 50 ($\square$), 55 ($\blacktriangledown$), and 60 ($\triangledown$)].

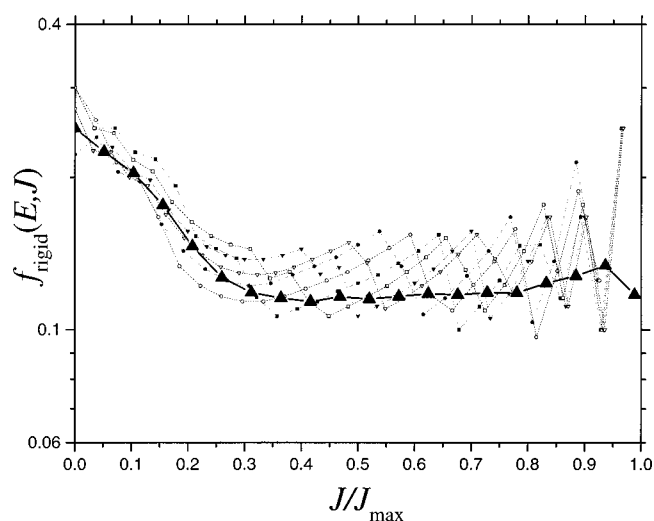


FIG. 21. As in Fig. 20; for $M = 0.001$, $J_{\text{max}} = 1.119(E/B)^{1/2}$, $E/B = 550$ ($\bullet$), 600 ($\circ$), 650 ($\blacksquare$), 700 ($\square$), 750 ($\blacktriangledown$), and 800 ($\triangledown$).

the present work we went one step further. We calculated axially nonadiabatic channel potential curves for atom+linear rotor-valence systems; from these we derived the corresponding specific rigidity factors and compared them with CT results. In doing such calculations, nonadiabaticity due to the kinetic energy part of the radial Hamiltonian was neglected while nonadiabaticity induced by the Coriolis coupling term of the Hamiltonian was included. Whenever large ranges of J values are relevant, the dominant part of nonadiabaticity of the dynamics stems from Coriolis coupling such as illustrated in the following.

Figure 19 shows examples of axially nonadiabatic channel potential curves (including Coriolis coupling) for atom+linear rotor curves in comparison to adiabatic channels (neglecting Coriolis coupling). An effective mass of $M = 0.02$ was chosen in Fig. 19 which, according to the results of Sec. VI (Fig. 14), corresponds to the intermediate range between adiabatic and sudden dynamics. The correlations¹⁰ between the orbital quantum numbers l and the projection quantum numbers Ω are well illustrated. A variety of compensating nonadiabatic radial coupling effects can be recognized in Fig. 19.

On average, the results from the explicit channel calculations fluctuate around the results from CT calculations. This is illustrated in Figs. 20 and 21 where specific rigidity factors $f_{\text{rigid}}(E, J)$ from our nonadiabatic channel calculations are represented together with the corresponding CT results. Figure 20 with an effective mass $M = 0.1$ according to Fig. 14 is only slightly nonadiabatic, whereas Fig. 21 is close to the completely nonadiabatic, sudden, limit. Because of the increasingly larger steps of $W^{\text{PST}}(E, J)$ with increasing J , the fluctuations of the nonadiabatic channel treatment are strongest for the largest J . Apart from these fluctuations, the agreement between (non)adiabatic channel and CT calculations convincingly documents the correspondence and agreement of our two dynamic treatments for the atom+linear rotor capture process.

VIII. CONCLUSIONS

The present work concludes our series of SACM/CT studies of the capture/dissociation dynamics in a number of model systems, from ion–dipole, ion–quadrupole, dipole–dipole to atom–linear rotor and linear rotor–linear rotor valence potentials. The present last article provides specific rigidity factors $f_{\text{rigid}}(E, J)$ whereas the earlier articles investigated thermal rigidity factors $f_{\text{rigid}}(T)$. Whereas the earlier work rationalized thermal capture (and thermal high pressure recombination or dissociation) rate constants, the present work provides a dynamically correct access to the specific rate constants $k(E, J)$ of unimolecular dissociation reactions. It is shown how $k(E, J)$ can be calculated in various steps, first by phase space theory neglecting centrifugal barriers $E_0(J)$, then by accounting for the centrifugal barriers $E_0(J)$ but still neglecting the anisotropy of the potential, and finally by also accounting for the anisotropy such as represented by the specific rigidity factors. Simple approximations for the various steps of the treatment are provided for the case that explicit information on the potential energy surfaces from *ab initio* calculations is not available.

Our treatment is fully dynamic in the capture process, covering the full range from adiabatic to nonadiabatic, sudden, dynamics. The analysis well documents, on the different stages of the treatment, the properties of “transitional modes” which are not conventional hindered rotors but “special modes,” governed by angular momentum couplings and a variety of complicated dynamical constraints. The combination of (non)adiabatic channel and classical trajectory calculations allows one to economically handle situations with only very few channels on the one hand or large numbers of channels on the other hand. In this way, special effects from open electronic shells can also be accounted for. Our treatment in many ways goes beyond variational transition state theory whose limitations remain to be investigated in detail.

ACKNOWLEDGMENT

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APPENDIX: THERMAL RIGIDITY FACTORS FOR LINEAR ROTOR+LINEAR ROTOR CAPTURE PROCESSES

In our previous work,⁵ we have only treated the case of the combination of two identical linear rotors forming adducts of various structures. In the following, we give the corresponding results for two nonidentical linear rotors of rotational constants B_1 and B_2 . Thermal rigidity factors $f_{\text{rigid}}(T)$ for low temperatures are again determined by CT calculations such as described in Ref. 5. They are well represented in the form

$$f_{\text{rigid}}(T \rightarrow 0) \approx (1 + 0.75Z + Z^4)^{-1/4} \quad (\text{A1})$$

with $Z = C^n/\gamma$. For the different types of adducts, the following values of C , n , and γ were derived.

Linear adduct: $C = \epsilon_1^2 \epsilon_2^2 / 4D^2 B_1 B_2$, $n = 2$, $\gamma = 2$, ϵ_1 and ϵ_2 are the quanta of the two doubly degenerate bending vibrations of the adduct.

T-shaped adduct: rotor 1 directed along the molecular axis, rotor 2 directed perpendicular to it, ϵ_1 and ϵ_2 are the quanta of the vibrations in the molecular plane, ϵ_0 is the quantum of the vibration perpendicular to the molecular plane, $C = (\epsilon_1^2 \epsilon_2^2 \epsilon_0^2 / B_1^2 B_2)^{1/3} / 2D$, $n = 3/2$, $\gamma = 6^{1/2}$; in the case of $\epsilon_0 \rightarrow 0$, $n \rightarrow 1$, and $\gamma \rightarrow 4.8$.

Adduct with parallel rotors: ϵ_t = quantum of torsional vibration, ϵ_s and ϵ_a = quanta for symmetric and asymmetric bending vibrations, $C = \{2\epsilon_s^2 \epsilon_a^2 \epsilon_t^2 / [B_1 B_2 (B_1 + B_2)]\}^{1/3} / 2D$, $n = 3/2$, $\gamma = 2(12)^{1/2}$ for symmetric rotors, $\gamma = 12^{1/4}$ for non-symmetric rotors; in the case of $\epsilon_t \rightarrow 0$, $n \rightarrow 1$, and $\gamma \rightarrow 4$.

Adduct with rotors of angle θ between both rotor axes and molecular axis, see Ref. 5: $\chi_a \approx \epsilon_a / \epsilon_t$, $\chi_s \approx \epsilon_s / \epsilon_t$, $n = 1 - 0.5 \sin^2 \theta + \sin^4 \theta$, $\gamma = d^{-n}$, $d = c_1 + c_2 \sin^2 \theta + c_3 / \sin^2 \theta + (\chi_a \chi_s)^{1/3} (c_4 + c_5 \sin^2 \theta + c_6 / \sin^2 \theta) + \chi_a^2 \cos^2 \theta (c_7 + c_8 \times \sin^2 \theta + c_9 / \sin^2 \theta) + \chi_s^2 \cos^2 \theta (c_{10} + c_{11} \sin^2 \theta + c_{12} / \sin^2 \theta)$ with $c_1 = 1.757$, $c_2 = -1.337$, $c_3 = -0.393$, $c_4 = 1.444$, $c_5 = -0.509$, $c_6 = -7.730$, $c_7 = 0.357$, $c_8 = -0.557$, $c_9 = 0.247$, $c_{10} = -1.932$, $c_{11} = 0.385$, $c_{12} = 1.529$; $C = \{2\epsilon_s^2 \epsilon_a^2 \epsilon_t^2 / [B_1 B_2 (B_1 + B_2)]\}^{1/3} / 2D$.

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